Designing Windows 365 as a PAW

Yes, your privileged access workstation can be a Cloud PC

Valtteri Aho | 21.10.2026 MSUG

Valtteri Aho

Vaasa

Team Lead; Endpoint management, Senior Consultant and Microsoft Certified Trainer

Context& Finland Oy, (former Sulava)

Over 19 years of experience as a trainer, system administrator and versatile helpdesk/datacenter expert. Over time, I have saved a lot of information about on-prem Microsoft technologies, Microsoft's Azure/M365 and Modern Device Management, not forgetting Config Manager.

Over 6 years at Sulava/Context&, focusing purely on Endpoint management modernization and development projects and of course, training. One of my current focus is to create AI based solutions for Endpoint management and security.



https://bsky.app/profile/wallopro.bsky.social

https://www.linkedin.com/in/valtteri-aho/

https://wallo.pro/

Why we still talk about PAWs

Admins are the shortest path to the crown jewels

- Attackers go after the people who hold the keys, not the servers.
- One admin session on an everyday laptop can undo every other control.
- Email, browsing and admin work on the same device is the classic risk.
- A PAW separates “doing admin work” from “doing everything else”.



What makes a workstation a PAW

The principles, independent of the technology

Dedicated

- Used only for admin work
- No email or everyday browsing

Hardened

- Minimal software
- Users cannot install or change things

Separate identity

- Admin account differs from daily account
- Strong, phishing-resistant sign-in

Enforced

- Admin tools reachable only from the PAW
- Activity is monitored

Why traditional PAWs often stall

Good idea, hard to live with

Hardware

- Extra laptops to buy, ship and replace
- Slow to deliver to new admins

User experience

- Two devices on the desk
- Admins find workarounds

Operations

- Separate builds to maintain
- Hard to scale beyond a few people

Partners

- External admins rarely get one
- Gaps where risk is highest

Enter Windows 365

A full Windows PC that lives in the Microsoft cloud

- Delivered quickly: no hardware logistics.
- Opens next to the daily desktop: one physical device, two separate worlds.
- Built from one standard image and policy set: every PAW looks the same.
- Easy to give to partners and external admins, and easy to take back.
- Removed or rebuilt quickly if something looks wrong.



A simple tiering model

Different blast radius, different level of protection

Tier 0	**On-premises identity** Active Directory, domain controllers, certificates, identity sync	**Highest isolation**
Tier 1	**Cloud control plane** Entra ID, Conditional Access, Intune, Defender, Azure subscriptions	**Privileged PAW**
Tier 2	**Workloads and business apps** Exchange, SharePoint, Teams, Power Platform, business systems	**Specialized PAW**
Tier 3	**User and device operations** Service desk, endpoint support, user lifecycle	**Compliant managed device**

Possibility 1: Cloud control plane (Tier 1)

Entra ID, Conditional Access, Intune, Defender

The Cloud PC

- Dedicated privileged profile
- No email, no everyday browsing
- Only approved admin sites and tools

The admin

- Separate admin account, no mailbox
- Roles activated only when needed
- Phishing-resistant sign-in

The rule

- Admin portals open only from the PAW
- The same sign-in from a normal laptop is blocked

Possibility 2: Microsoft 365 workloads (Tier 2)

Exchange, SharePoint, Teams, Power Platform

The Cloud PC

- Specialized profile
- Can be shared by several workload admins
- Email and browsing can be allowed

The admin

- Separate admin account
- Just-in-time role activation recommended

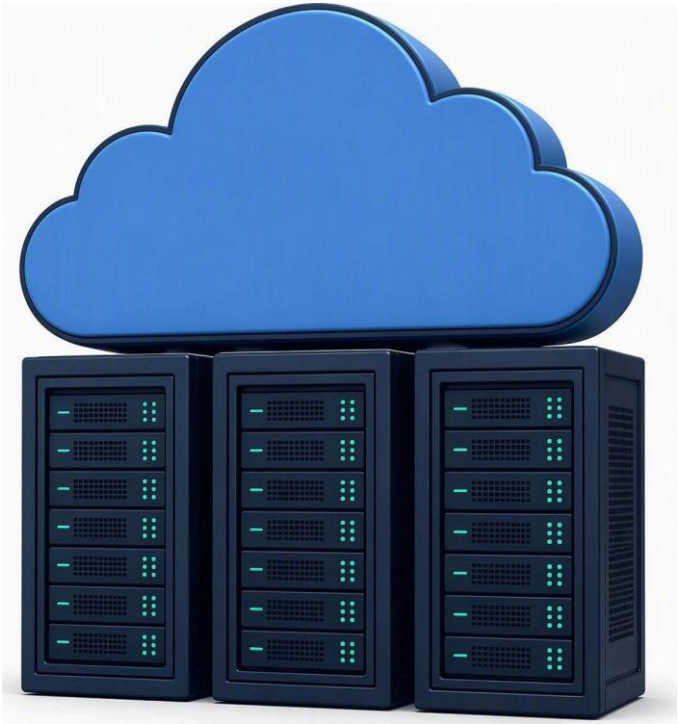
The value

- Business-data admin isolated from daily work
- Same standard build as Tier 1, slightly relaxed

Possibility 3: Azure administration (Tier 2)

Subscriptions, networks, workloads

- Azure portal and admin tools allowed only from the PAW.
- Subscription-wide owners belong with the control plane (Tier 1).
- Application and resource admins can use a specialized profile.
- Cloud PCs can be connected close to the Azure resources they manage.
- Consistent rules for internal admins, DevOps teams and partners.



Possibility 4: On-premises and hybrid (Tier 0)

A Cloud PC can reach your datacenter too

How it works

- Cloud PC connected to your own network
- Admin tools reach servers as from a jump host
- Can replace many legacy jump servers

Good fit

- Server and application administration
- Hybrid environments in transition
- Tier 1 and Tier 2 on-premises systems

Think carefully

- Tier 0: Active Directory and identity
- Requires a separate, dedicated design
- Some organisations keep physical PAWs here

One model, a Cloud PC per tier

Possibilities across the whole estate

Tier	Scope	Windows 365 option	What is allowed
Tier 0	On-premises identity	Dedicated Cloud PC or physical PAW	Admin tools only
Tier 1	Cloud control plane and Azure owners	Dedicated privileged Cloud PC	Approved admin sites only
Tier 2	M365 workloads and business apps	Specialized Cloud PC, can be shared	Admin work plus email and browsing
Tier 3	User and device operations	Not required	Compliant managed device

Never mix tiers on the same Cloud PC. Higher tier = stricter rules.

The honest part: the device in front of you

A Cloud PC is only as trustworthy as the way you reach it

- The access device should be managed and healthy, not a personal laptop.
- Higher tiers need a cleaner access device: a hardened laptop or thin client.
- Sign-in to the Cloud PC must be phishing-resistant.
- Restrict copy-paste, file transfer and printing out of the PAW.
- Watch the admins: sign-ins, role activations and Cloud PC activity.



Guardrails that make it work

In plain words

Who

- Separate admin accounts
- Roles only when needed, time-limited

How

- Phishing-resistant sign-in
- Admin portals only from the PAW

What

- Minimal apps, no local admin rights
- Only approved sites and tools

Watch

- Central monitoring
- Rebuild instead of repair

Windows 365 delivery models for PAWs

Separate machines or shared machines? These are different decisions.

Enterprise personal

One Cloud PC per admin

Persistent environment

Recommendation: regular, high-impact privileged work

Flex dedicated

Up to 3 separate PCs per license

1 concurrent session per license

Persistent, user-assigned PCs

Recommendation: occasional admins

Flex shared

1 PC per license

Group uses it sequentially

1 active user per PC

Recommendation: narrowly scoped tasks

Recommendation: share capacity before sharing the workstation.

Shared PAWs: trade-offs and safeguards

Shared does not mean multi-session, and profile cleanup is not an OS rebuild.

What is shared?

Different admins reuse the OS

Only one user at a time

Keep each pool within one trust boundary

What persists?

Profiles deleted at sign-out by default

User Experience Sync can preserve settings and app data

Recommendation: disable sync for a nonpersistent design

What to plan?

Shared has no concurrency buffer

Size for incident-time demand

Restrict tools and local privileges

Monitor and reprovision as needed

Recommendation: dedicated PCs for identity and cloud-control-plane admins.

What about Azure Virtual Desktop as a PAW?

Both can work; they suit different organisations

	Windows 365	Azure Virtual Desktop
What you get	A ready Cloud PC per admin	A platform you build and run
Effort to run	Low, largely managed by Microsoft	Higher, your team manages the hosts
Cost model	Fixed monthly price per admin	Pay for capacity used
Isolation	Personal Cloud PC per admin	Personal or shared hosts, by design
Flexibility	Standard sizes and options	Full control of size, network, scaling
Best fit	Most admin teams, fast start	Specialised needs, existing AVD skills

Key takeaways

- PAW principles stay the same; only the delivery changes.
- Windows 365 fits Tier 1, Tier 2, Azure and on-premises administration.
- Tier 0 needs a conscious decision: Cloud PC or physical PAW.
- Trust in the access device and sign-in still matters.
- Start with a small control-plane pilot and grow by tier.

Questions?

Valtteri Aho | Thank you

https://bsky.app/profile/wallopro.bsky.social

https://www.linkedin.com/in/valtteri-aho/

https://wallo.pro/